

Enterprise Digital Infrastructure – Self-assessment: Page Load Time

April 29, 2026

First name/Lastname _____

ID _____

- A. Consider a web page consisting of an HTML file (2Kbytes), two JavaScripts (4Kbyte each), two style sheets (8Kbytes each) and three images (16Kbytes each). The browser is running on a client in the domain **unipv.it**. The page and all referenced objects are stored on the server **www.abc.io**.

Assumptions: RTT between the client and the web server: 10 msec. Bandwidth between the client and the web server: 8Mbps. Persistent TCP connection between the client and the web server. No parallel connections.

- a. Compute the number of lookup phases and for each lookup specify the Name Server involved, the number of RTTs and your assumptions, if any.

- b. Compute the Page Load Time (write the expression and make the computations) under the assumption of an unsecure connection.

- B. Consider a web page consisting of an HTML file, one JavaScript, two style sheets and one image. The browser is running on a client in the domain **domain.com**. The HTML file and the JavaScript are stored on the server **www.abc.io**. The style sheets are stored on the server **css.abc.com**, the image on the server **img.abc.io**.

- a. Compute the number of lookup phases and for each lookup specify the Name Server involved, the number of RTTs and your assumptions, if any.

- b. Compute the number of RTTs requested for downloading the entire web page under the assumption of an unsecure connection.

 - c. Compute the number of RTTs requested for downloading the entire web page under the assumption of a secure connection.
- C. Consider the download of a web page using a secure connection. The page includes 100 referenced objects, each stored on a different web server.
- a. Compute the number of TCP connections opened by the browser.

 - b. Compute the number of TLS connections opened by the browser.

 - c. Compute the number of RTTs for downloading the entire page.